Daily Algebra Drill #6

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Comparing coefficients · Substitution $t = \sqrt{x}$ for surd equations · Schur's inequality · Integer-valued polynomials.

Section A Rapid Recognition

1. Factorise: $x^3 + 5x^2 + 6x$.

Ans: $x(x+2)(x+3)$

2. Given $x + \dfrac{1}{x} = \sqrt{5}$, find $x^2 + \dfrac{1}{x^2}$.

Ans: 3

3. State the coefficient of xy^2 in $(x+y)^3$.

Ans: 3

4. Evaluate $\dfrac{1}{\sqrt{2}-1} - \dfrac{1}{\sqrt{2}+1}$.

Ans: 2

5. If $x = 2$ is a repeated root of $f(x) = x^3 + ax + b$, find a and b.

Ans: $a = -12,\ b = 16$

6. Simplify: $\left(\dfrac{2x}{y}\right)^3 \div \dfrac{4x^2}{y^2}$.

Ans: $\dfrac{2x}{y}$

7. Given $\alpha\beta = 6$ and $\alpha + \beta = 5$, find $\alpha^2\beta + \alpha\beta^2$.

Ans: 30

8. Factorise: $4a^2 - 4ab + b^2$.

Ans: $(2a - b)^2$

9. Simplify $2^{10} - 1$ as a product of three factors greater than 1.

Ans: $3 \times 11 \times 31$

10. Given $f(x) = x^2 - 3x + 2$, find $f(f(0))$.

Ans: 0

Section B Manipulation Drills

1. Find A, B, C such that $x^2 + 3x + 5 \equiv A(x+1)^2 + B(x+1) + C$.

Ans: $A = 1,\ B = 1,\ C = 3$

2. The cubic $p(x) = x^3 + ax^2 + bx + c$ satisfies $p(1) = 0$, $p(-1) = 4$, $p(2) = 15$. Find a, b, c.

Ans: $a = \frac{11}{3}$, $b = -3$, $c = -\frac{5}{3}$

3. Simplify: $\dfrac{x^3 - 8}{x^2 + 2x + 4}$.

Ans: $x - 2$

4. State the identity $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ and verify at $a = b = c = 1$.

Ans: At $a = b = c = 1$: LHS $= 3 - 3 = 0$; RHS $= 3(3 - 3) = 0$. ✓

5. Given $a+b+c=0$, simplify $\dfrac{a^3+b^3+c^3}{abc}$.

Ans: 3

6. Verify $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ for $n=1,2,3$; hence find $\sum_{k=1}^{10}(2k-1)^2$.

Ans: $\sum_{k=1}^{10}(2k-1)^2 = 1330$

7. Simplify: $\dfrac{\frac{1}{a+b}-\frac{1}{a-b}}{\frac{1}{a+b}+\frac{1}{a-b}}$.

Ans: $\frac{-b}{a}$

8. Make b the subject of $\dfrac{a-b}{a+b} = \dfrac{c-d}{c+d}$.

Ans: $b = \frac{ad}{c}$

9. Given α, β, γ are roots of $x^3 - 7x + 6 = 0$, find $\alpha^2 + \beta^2 + \gamma^2$ and $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$.

Ans: $\alpha^2 + \beta^2 + \gamma^2 = 14$; $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = 49$

10. Solve: $\dfrac{2}{x^2-1} - \dfrac{1}{x-1} = \dfrac{3}{x+1}$.

Ans: No solution (the only candidate $x = 1$ is excluded from the domain).

Section C Substitution & Structure

1. Solve: $\sqrt{x} - \dfrac{6}{\sqrt{x}} = 1$.

Ans: $x = 9$

2. Find the minimum of $x^2 - 6x + y^2 + 4y + 20$.

Ans: Minimum is 7, at $(x, y) = (3, -2)$.

3. Show $a^4 + b^4 \geq \frac{1}{2}(a+b)^4$ is false; prove the correct inequality $a^4 + b^4 \geq \frac{1}{2}(a^2+b^2)^2$.

Ans: Counterexample: $a = b = 1$: $2 \geq \frac{1}{2}(16) = 8$? False. Correct inequality: $(a^2-b^2)^2 \geq 0 \Rightarrow a^4 + b^4 \geq 2a^2b^2 \Rightarrow 2(a^4+b^4) \geq (a^2+b^2)^2$.

4. Solve $x^4 - 2x^3 - 3x^2 + 4x + 4 = 0$ given that it has a repeated real root.

Ans: $x = 2$ (double root) and $x = -1$ (double root)

5. Given $a, b, c > 0$ with $a + b + c = 1$, prove $ab + bc + ca \leq \frac{1}{3}$.

Ans: Proved below.

6. If $x + y = 1$, find the maximum of $x^2y + xy^2$.

Ans: $\frac{1}{4}$

7. Solve over $\mathbb{R}$: $\sqrt{3x+1} - \sqrt{x+4} = 1$.

Ans: $x = 5$ only

8. For what values of b, c do $x^2 + bx + c = 0$ and $x^2 + cx + b = 0$ have the same roots?

Ans: $b = c$

Section D Challenge (SMC / BMO1 difficulty)

1. Schur's Inequality (degree 2): Prove $\sum a^2(a-b)(a-c) \geq 0$ for $a, b, c \geq 0$.

Ans: Proved below.

2. Find all polynomials with real coefficients satisfying $p(x^2) = p(x)^2$.

Ans: $p(x) = 0$ and $p(x) = x^n$ for non-negative integers n.

3. Find a closed form for $\dfrac{1^2}{1\cdot 3}+\dfrac{2^2}{3\cdot 5}+\cdots+\dfrac{n^2}{(2n-1)(2n+1)}$.

Ans: $\dfrac{n(n+1)}{2(2n+1)}$

4. A polynomial $p(x)$ of degree 3 satisfies $p(n)=\frac{1}{n}$ for $n=1,2,3,4$. Find $p(5)$.

Ans: $p(5)=0$

5. Find all pairs of positive integers (m, n) with $m > n$ such that $m^2 - n^2 = 105$.

Ans: $(m, n) \in \{(53, 52), (19, 16), (13, 8), (11, 4)\}$